

1 An exact low-height exclusion from the original theta boundary

This calculation keeps the original boundary $f_0 = \Theta\phi_*$, its Mellin transform $g = 2\xi$, and its unscaled positive kernel ψ . In particular, the constant $g(1) = 1$ below is a value calculated from that boundary, not a normalization imposed on g . Every estimate is an inequality of actual integrals. The numerical lower bound at the end is a rational number proved positive by exact arithmetic.

1.1 The original source and its exact endpoint mass

Use the functions and coordinates in (HCA.1)–(HCA.2):

$$\begin{aligned}\vartheta(x) &= \sum_{n \in \mathbb{Z}} e^{-\pi n^2 x^2} = 1 + 2 \sum_{n \geq 1} e^{-\pi n^2 x^2}, \\ \phi_*(x) &= (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \\ f_0(x) &= \Theta\phi_*(x) = 2 \sum_{n \geq 1} \phi_*(nx), \\ \psi(y) &= e^{y/2} f_0(e^y), \\ g(s) &= \mathcal{M}f_0(s) = \int_0^\infty f_0(x) x^s \frac{dx}{x} = \int_{\mathbb{R}} e^{(s-1/2)y} \psi(y) dy.\end{aligned}\tag{LHT.1}$$

Here $x > 0$ and $y = \log x \in \mathbb{R}$. The original theta identity is $\vartheta(x) = x^{-1}\vartheta(x^{-1})$. This is the same Gaussian theta identity used in the source proof of $\psi(-y) = \psi(y)$ in (HCA.1); no different theta function is introduced here. Its series and every differentiated series converge uniformly on compact subsets of $x > 0$. Differentiating them gives

$$\begin{aligned}\vartheta'(x) &= -4\pi x \sum_{n \geq 1} n^2 e^{-\pi n^2 x^2}, \\ (x^2 \vartheta'(x))' &= 8\pi^2 x^4 \sum_{n \geq 1} n^4 e^{-\pi n^2 x^2} - 12\pi x^2 \sum_{n \geq 1} n^2 e^{-\pi n^2 x^2} = f_0(x).\end{aligned}\tag{LHT.2}$$

For $x \geq 1$, every term of $f_0(x)$ is positive, since $4\pi^2 n^4 x^4 - 6\pi n^2 x^2 = 2\pi n^2 x^2(2\pi n^2 x^2 - 3) > 0$. The evenness of ψ therefore gives $\psi(y) > 0$ for every real y . On $y \geq 0$ we retain the termwise upper bound

$$0 < \psi(y) \leq 8\pi^2 \sum_{n \geq 1} n^4 \exp\left(\frac{9}{2}y - \pi n^2 e^{2y}\right).\tag{LHT.3}$$

Together with evenness, this is an integrable majorant after multiplication by any fixed polynomial in y and any fixed $e^{|y|}$. For example, for $y \geq 0$ the sum is bounded by $(\sum_{n \geq 1} n^4 e^{-\pi(n^2-1)})e^{9y/2-\pi e^{2y}}$ times $8\pi^2$. All Mellin integrals and differentiations below consequently converge absolutely; termwise upper bounds on their nonnegative integrands also follow directly from monotone convergence.

The limits in (LHT.2) can be evaluated before integrating. At $x \rightarrow \infty$, the differentiated theta series gives $x^2 \vartheta'(x) \rightarrow 0$. Differentiating the theta identity itself gives the exact equation

$$x^2 \vartheta'(x) = -\vartheta(x^{-1}) - x^{-1} \vartheta'(x^{-1}).\tag{LHT.4}$$

As $x \rightarrow 0^+$ the two terms on the right tend to -1 and 0 , respectively. Integrating (LHT.2) first on a compact interval and then taking its endpoints to 0 and ∞ is justified by the absolute convergence just proved. It follows that

$$g(1) = \int_0^\infty f_0(x) dx = [x^2 \vartheta'(x)]_{0^+}^\infty = 1.\tag{LHT.5}$$

This calculation does not interchange the integrals of the signed individual theta terms on the whole positive axis.

Define two actual, unscaled source moments

$$\begin{aligned} M_0 &= \int_{\mathbb{R}} \psi(y) dy = g(1/2), \\ G_2 &= \int_{\mathbb{R}} y^2 e^{y/2} \psi(y) dy = 2 \int_0^\infty y^2 \cosh(y/2) \psi(y) dy = g''(1). \end{aligned} \tag{LHT.6}$$

Both numbers are strictly positive. For real z , $\cosh z - 1 \leq \frac{1}{2}z^2 \cosh z$: for $z \geq 0$ this follows from $\cosh z - 1 = \int_0^z (z-v) \cosh v dv \leq \cosh z \int_0^z (z-v) dv$, and both sides are even. Equations (LHT.5)–(LHT.6) therefore imply

$$1 - M_0 = 2 \int_0^\infty (\cosh(y/2) - 1) \psi(y) dy \leq \frac{1}{8} G_2, \quad M_0 \geq 1 - \frac{1}{8} G_2. \tag{LHT.7}$$

1.2 A rational upper bound on the unscaled second moment

We now bound G_2 without a floating-point computation or a zero-location theorem. For $y \geq 0$ use $\cosh(y/2) \leq e^{y/2}$ in (LHT.3), and then put $x = e^y$. The resulting nonnegative sum and integrals give

$$\begin{aligned} G_2 &\leq 16\pi^2 \sum_{n \geq 1} n^4 \int_0^\infty y^2 e^{5y - \pi n^2 e^{2y}} dy \\ &= 16\pi^2 \sum_{n \geq 1} n^4 \int_1^\infty (\log x)^2 x^4 e^{-\pi n^2 x^2} dx. \end{aligned} \tag{LHT.8}$$

For $x \geq 1$, $\log x \leq x - 1$. Set $v = x - 1 \geq 0$ and use $(1+v)^2 \geq 1 + 2v$. Expanding the remaining polynomial, with no change to the coordinate or the original moment, yields

$$\begin{aligned} G_2 &\leq 16\pi^2 \sum_{n \geq 1} n^4 e^{-\pi n^2} \int_0^\infty v^2 (1+v)^4 e^{-2\pi n^2 v} dv \\ &= \sum_{n \geq 1} e^{-\pi n^2} \sum_{r=0}^4 \frac{16 \binom{4}{r} (r+2)!}{2^{r+3} \pi^{r+1} n^{2r+2}}. \end{aligned} \tag{LHT.9}$$

The integral identity used here is $\int_0^\infty v^j e^{-av} dv = j! a^{-j-1}$ for $a > 0$ and nonnegative integer j ; it follows by j integrations by parts, with base integral $1/a$ and zero endpoint terms.

The elementary inequality $\pi > 3$ and $n \geq 1$ bound the five terms of the inner sum, respectively, by

$$\frac{4}{3}, \quad \frac{8}{3}, \quad \frac{8}{3}, \quad \frac{40}{27}, \quad \frac{10}{27}; \quad \frac{4}{3} + \frac{8}{3} + \frac{8}{3} + \frac{40}{27} + \frac{10}{27} = \frac{230}{27}. \tag{LHT.10}$$

For the exponential bound, the exact finite Taylor sum gives

$$e^3 > \sum_{j=0}^8 \frac{3^j}{j!} = \frac{89641}{4480} > \frac{89600}{4480} = 20. \tag{LHT.11}$$

Thus $e^{-\pi n^2} < 20^{-n^2} \leq 20^{-n}$ for $n \geq 1$, and

$$\sum_{n \geq 1} e^{-\pi n^2} < \sum_{n \geq 1} 20^{-n} = \frac{1}{19}. \tag{LHT.12}$$

Combining (LHT.9)–(LHT.12) proves the entirely rational bound

$$0 < G_2 = g''(1) < \frac{230}{27 \cdot 19} = \frac{230}{513}. \tag{LHT.13}$$

In particular this proof includes every $n \geq 1$ and the entire positive integration axis. There is no unestimated theta tail.

1.3 Uniform exclusion in the closed low-height rectangle

Write $s = \frac{1}{2} + a + it$, where $a, t \in \mathbb{R}$. Evenness and reality of ψ in (LHT.1) give

$$\operatorname{Re} g(\tfrac{1}{2} + a + it) = 2 \int_0^\infty \psi(y) \cosh(ay) \cos(ty) dy. \quad (\text{LHT.14})$$

The real inequality $\cos z \geq 1 - z^2/2$ follows from $1 - \cos z = 2 \sin^2(z/2) \leq z^2/2$. If $|a| \leq 1/2$, then $1 \leq \cosh(ay) \leq \cosh(y/2)$ for $y \geq 0$. Since every factor $\psi(y) \cosh(ay)$ is nonnegative, substitution of this cosine inequality in (LHT.14) gives

$$\begin{aligned} \operatorname{Re} g(\tfrac{1}{2} + a + it) &\geq 2 \int_0^\infty \psi(y) \cosh(ay) dy - t^2 \int_0^\infty y^2 \psi(y) \cosh(ay) dy \\ &\geq M_0 - \frac{t^2}{2} G_2. \end{aligned} \quad (\text{LHT.15})$$

For $|t| \leq 2$, equations (LHT.7) and (LHT.13) now imply

$$\boxed{\operatorname{Re} g(s) \geq 1 - \frac{17}{8} G_2 > 1 - \frac{17}{8} \frac{230}{513} = \frac{97}{2052} > 0, \quad 0 \leq \operatorname{Re} s \leq 1, \quad |\operatorname{Im} s| \leq 2.} \quad (\text{LHT.16})$$

Consequently $g(s) = 2\xi(s)$ has no zero in that closed rectangle. This is proved directly from the actual theta source, without assuming the Riemann hypothesis or using numerical verification of its low zeros.

The resulting map on the programme's original local packet objects is also explicit. At every point s_0 of this rectangle, $g(s_0) \neq 0$, so multiplication by g on the local holomorphic ring $\mathcal{O}_{\mathbb{C}, s_0}$ is an isomorphism with inverse multiplication by the holomorphic germ $1/g$. On the rectangle itself the preceding calculation proves

$$|1/g(s)| \leq \frac{1}{\operatorname{Re} g(s)} < \frac{2052}{97}. \quad (\text{LHT.17})$$

It follows that $\mathcal{O}_{\mathbb{C}, s_0}/(g) = 0$ at every such s_0 . Under the programme's already constructed Mellin-jet isomorphism to the actual packet summands, each of these local summands therefore vanishes. No proper-source kernel is quotiented out by this argument. In particular an actual upper-half-plane offcritical centre obeys

$$g(\rho) = 0, \quad \rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 0 \implies \gamma > 2. \quad (\text{LHT.18})$$

Its reflected and conjugate centres retain their original orders, jets, and Taylor units.